

Breast Cancer Classification using Feedforward Neural Network

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Github Link: <https://github.com/Srinitya22/Breast-Cancer-Classification-FFNN>

Abstract

Breast cancer is a disease that threatens the lives of women all over the world. It is among the diseases that causes the highest rates of cancer-related deaths. Early diagnosis increases the likelihood of successful treatment to cure breast cancer. AI plays a critical role in diagnosis. This project proposes a Feedforward Neural Network model for classifying breast cancer tumors into Malignant and Benign categories. The ML model was tested using Wisconsin Diagnostic Breast Cancer dataset. Besides using some of the best techniques to improve generalization and reduce overfitting, combination of different optimizers and loss functions were tested. The resulting model had an accuracy of 100 percent on the test set.

1. Introduction

Breast cancer is a typical disease affecting many people worldwide. Diagnosis plays a critical role in treating this disease. Traditional methods of diagnosis require professional evaluation by lab tests which may take time. Artificial Intelligence methods such as Deep Learning are capable of analyzing patient data to make accurate predictions.

FNN is one of the most efficient NN architectures that is suitable for solving classification problems. This project involves developing an FNN for the categorization of breast cancer tumors.

2. Objective

This study aims to:

- Develops a Feedforward Neural Network for categorizing breast cancer.
- Standardize data before model training and reduce overfitting using some of the best techniques.
- Compare different optimizers and loss functions.
- Evaluate the final model using classification metrics.

3. Dataset Description

The Wisconsin Diagnostic Breast Cancer dataset was used for this study. The dataset consists of 569 samples with 30 features. Only 20 of the features were used for model training since the other 10 features were not relevant for making predictions. The target variable consisted of two classes; Benign and Malignant with 357 and 212 samples respectively.

The sample distribution is shown below:

Benign: 357

Malignant: 212

The given data was partitioned as train, test sets, and validation with 398, 86, and 85 samples, respectively. Finally, StandardScaler() was used to scale the features using the training set fit in the scaler.

4. Methodology

The methodology followed in this study is given below:

- Dataset acquisition
- Feature selection
- Data Preprocessing
- Developing a train–validation–test split
- Feedforward Neural Network development
- Model Training
- Regularization
- Performance Evaluation

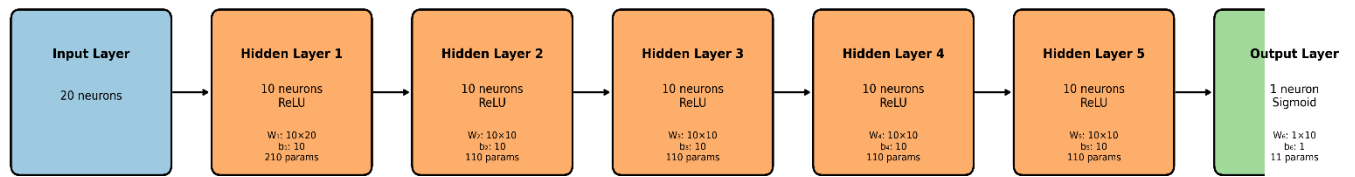
5. Neural Network Architecture

The baseline Feedforward Neural Network architecture has 20 neurons in the input layer, 10 neurons in each of the five hidden layers, and one neuron in the output layer for binary classification. Thus, the overall network architecture was $20 \rightarrow 10 \rightarrow 10 \rightarrow 10 \rightarrow 10 \rightarrow 10 \rightarrow 1$. The network has a total of 661 parameters as shown below:

- Input layer: 20 neurons
- Hidden layer 1: 10 neurons
- Hidden layer 2: 10 neurons
- Hidden layer 3: 10 neurons
- Hidden layer 4: 10 neurons
- Hidden layer 5: 10 neurons
- Output layer: 1 neuron

Feedforward Neural Network for Breast Cancer Classification

Architecture: $20 \rightarrow 10 \rightarrow 10 \rightarrow 10 \rightarrow 10 \rightarrow 10 \rightarrow 1$



Baseline total trainable parameters = 661 | Binary Cross-Entropy loss | Adam optimizer

Regularized version: Batch Normalization + Dropout($p=0.2$) + Early Stopping

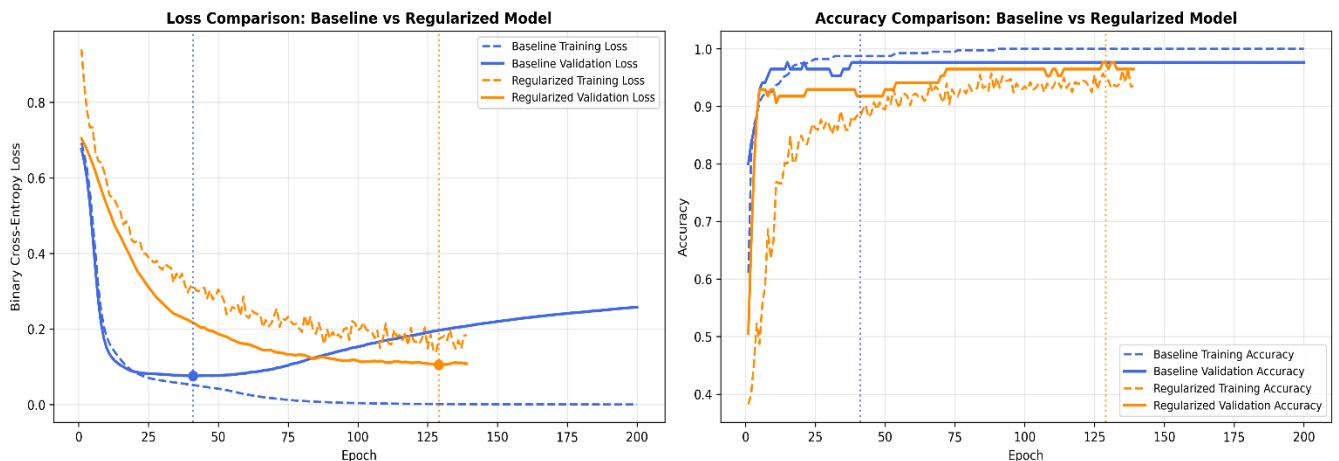
6. Regularization Techniques

The baseline model was found to be overfitting after training. Some of the regularization techniques used to reduce overfitting were:

- Batch normalization
- Dropout
- Early stopping
- Adam optimizer
- Binary cross-entropy loss function

The regularized network architecture followed the dense $\rightarrow$ batch normalization $\rightarrow$ ReLU activation $\rightarrow$ dropout pattern.

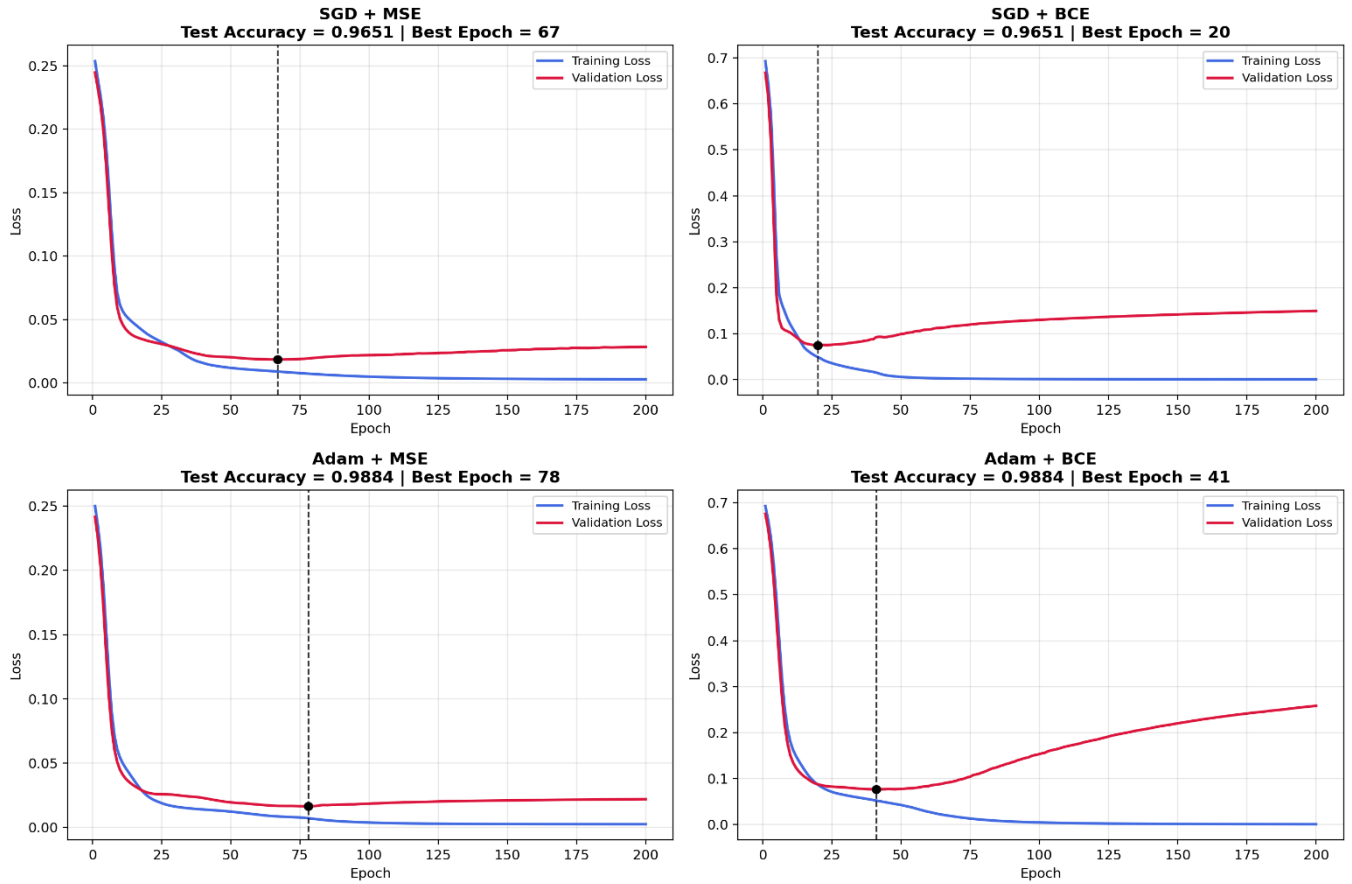
Effect of Batch Normalization, Dropout, and Early Stopping



7. Optimizer and Loss Function Comparison

SGD and Adam optimizer were compared using MSE and binary cross-entropy loss function for best performing model. The Adam optimizer was found to be the best optimizer among the rest.

2 × 2 Comparative Analysis: Optimizers × Loss Functions



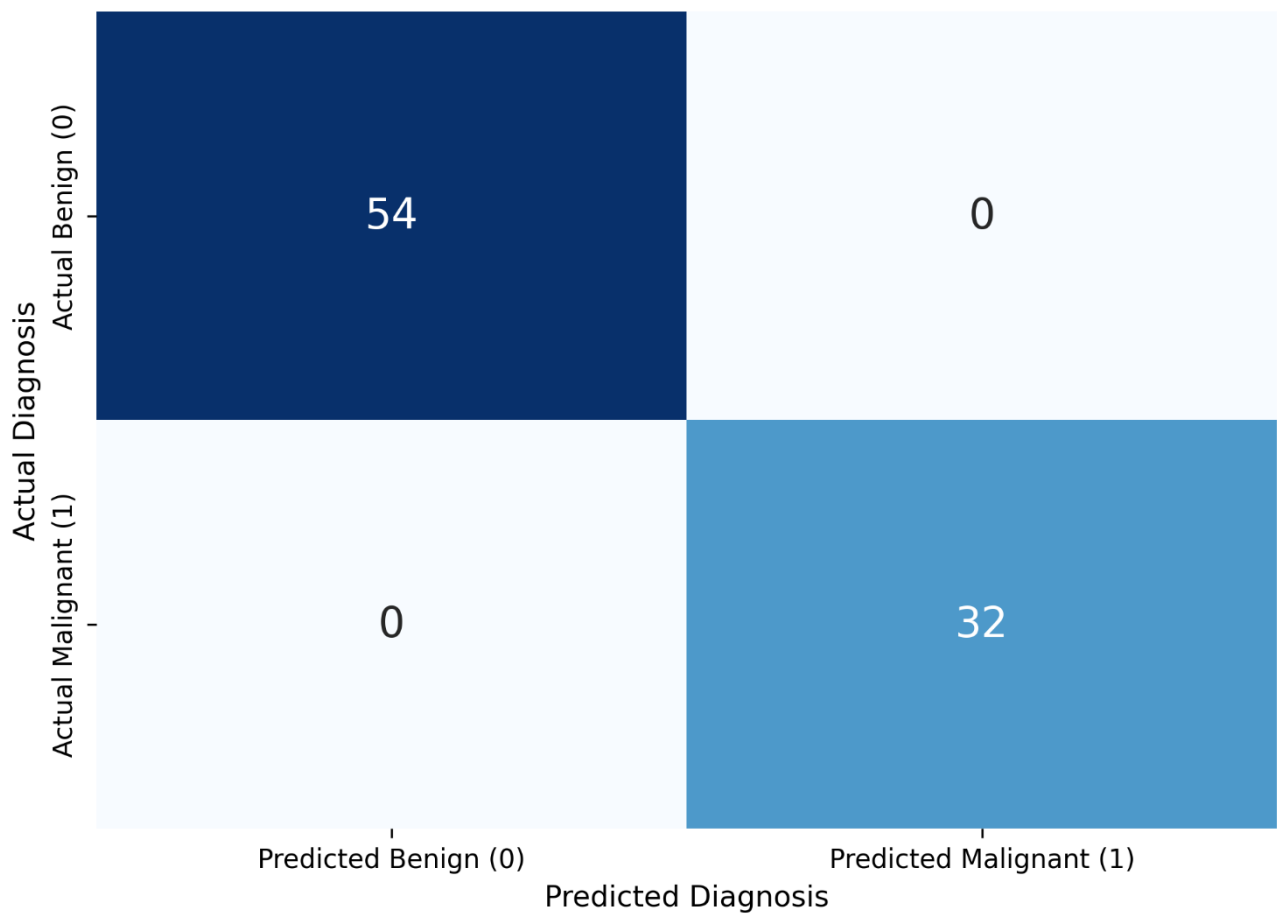
8. Results

The final regularized model was evaluated using the held-out test set with excellent results as shown:

- Test Accuracy: 100%
- Precision: 100%
- Recall: 100%
- Specificity: 100%
- F1-Score: 1.00
- AUC-ROC: 1.00

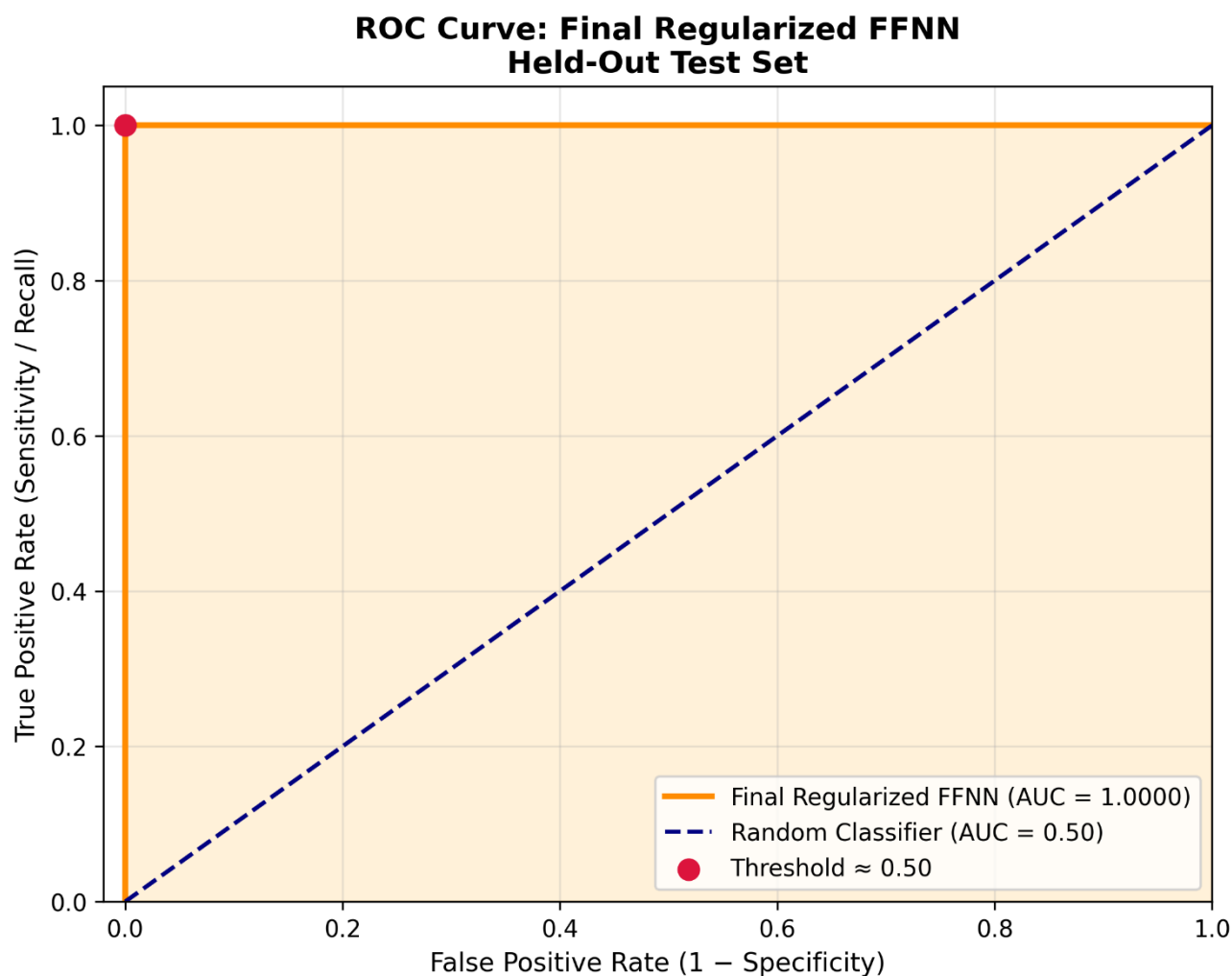
The model predicted all test samples correctly without any positive or false negatives.

**Confusion Matrix: Final Regularized FFNN
Held-Out Test Set**



9. ROC Curve

The receiver operating characteristic curve is a commonly utilized metric for evaluating classification models. The ROC curve for the proposed model is shown below. The area under the ROC curve was 1. This implies that the proposed model is able to distinguish between the two classes of tumors.



10.Applications

The proposed model can be applied in various fields, including:

- Breast cancer diagnosis
- decision support systems
- Medical image and healthcare analytics
- AI-assisted cancer screening
- Hospital diagnostic applications

11.Advantages

- Classification accuracy
- Effective reduction of overfitting
- Stable model convergence
- Simple Feedforward Neural Network architecture with minimal computational cost
- Appropriate for binary medical classification problems

12. Limitations

Despite the many advantages of using the proposed approach for tumor classification, it has various disadvantages, including:

- Small size of the dataset with only 569 samples.

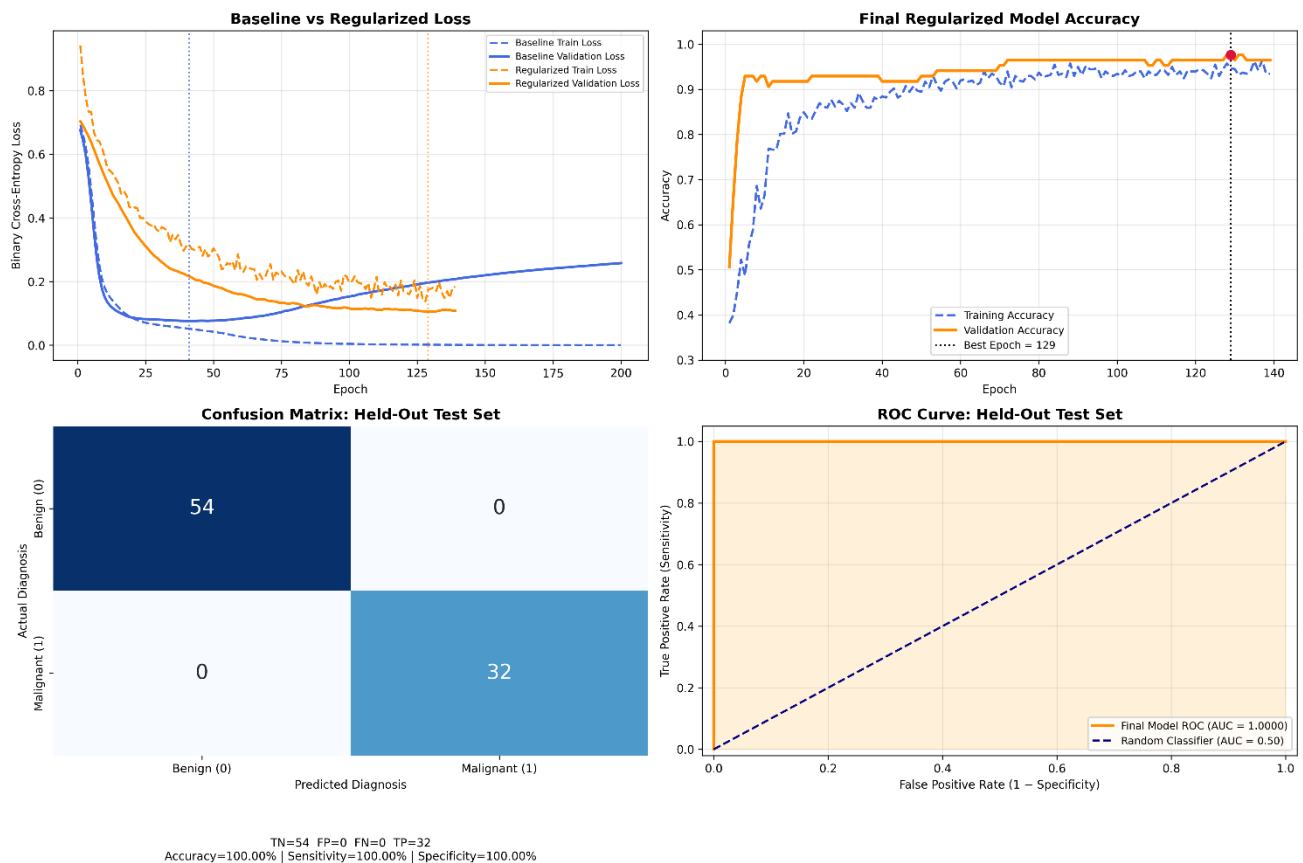
The model was evaluated using the train-validation-test split. Further research is needed to assess the performance of the proposed model using -validation folds & external datasets before applying it in real-world medical diagnostic applications.

13. Conclusion & Future Direction

This project proposed a successful Feedforward Neural Network model for classifying breast cancer tumors. The baseline model had high variance which was reduced using some of the best regularization techniques. The final regularized model perfectly classified the held-out test set with an accuracy of 100%, precision of 100%, recall of 100%, and AUC-ROC score of 1. The results show that deep learning approaches can be used to classify tumors after proper preprocessing and regularization techniques have been used.

14. Visual Dashboard

Breast Cancer FFNN Diagnostic Dashboard
20 Inputs → 5 Hidden Layers × 10 Neurons → 1 Output



References

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